
Averaging Lowers Memorization in DP-FedAvg: A Self-Contained Conditional-Mutual-Information Analysis

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Abstract

We analyze per-client memorization in differentially private federated averaging (DP-FedAvg) using the conditional mutual-information measure of Brown et al. [2021]. For one round of DP-FedAvg—per-client clipping at norm C , uniform averaging across K clients, and additive Gaussian noise of scale σ —we prove the explicit bound $I(X_k; \tilde{\Delta} \mid P) \leq C^2/(2K^2\sigma^2)$. The argument is fully elementary: we apply a single Gaussian-channel mutual-information inequality (Lemma 1) directly, bypassing any black-box “DP $\Rightarrow$ MI” reduction. We then extend to a T -round adaptive transcript via a telescoping chain-rule argument that correctly handles the explaining-away dependences induced by past releases, obtaining $I(X_k; \mathcal{T} \mid P) \leq \sum_t C^2/(2K^2\sigma_t^2)$. The per-client bound is sharper than the whole-dataset bound $I(X; \tilde{\Delta} \mid P) \leq C^2/(2\sigma^2)$, and than the analogous bound when each client’s clipped+noised update is released separately, by a factor of K^2 in both cases. Both bounds match the metric studied by Brown et al. and quantify how averaging—not noise alone—suppresses individual-client memorization in DP-FedAvg.

1 Introduction

In federated learning a central server coordinates training across K clients without collecting raw data. The canonical *DP-FedAvg* algorithm of McMahan et al. [2018] (see also Geyer et al. 2017) proceeds round-by-round: each client returns a clipped update, the server averages the clipped updates, adds Gaussian noise, and broadcasts the new iterate. This is the federated counterpart of central DP-SGD [Abadi et al., 2016, Dwork and Roth, 2014].

Brown et al. [2021] introduced an information-theoretic metric of memorization,

$$\text{Mem}(A; q) = I(X; A(X) \mid P),$$

where $P \sim q$ is a problem instance drawn from a meta-distribution q , $X \sim P^{\otimes n}$ is the dataset, and $A(X)$ is the model. They showed that, on natural learning tasks, every sufficiently accurate algorithm must satisfy $I(X; A(X) \mid P) = \Omega(nd)$. Their definition is precisely the conditional mutual information given the data distribution, not a per-record or worst-case quantity (cf. Brown et al., 2021, Theorem 1.1, Section 1.1).

Translating this metric to the federated setting raises a natural question: in DP-FedAvg, how does the per-client memorization $I(X_k; M \mid P)$ depend on K , the clip bound C , and the noise scale σ ?

Contributions. We give a fully explicit answer:

- **One-round bound (Theorem 3).** For one round of DP-FedAvg, $I(X_k; \tilde{\Delta} \mid P) \leq C^2/(2K^2\sigma^2)$. The proof uses only the Gaussian differential-entropy maximization inequality together with the data-processing inequality, and we make every conditional-independence step explicit.
- **Multi-round bound (Theorem 5).** For a T -round adaptive transcript $\mathcal{T} = (\tilde{\Delta}^{(0)}, \dots, \tilde{\Delta}^{(T-1)})$, $I(X_k; \mathcal{T} \mid P) \leq \sum_{t=0}^{T-1} C^2/(2K^2\sigma_t^2)$. Section 5.2 explains why the naive “apply Theorem 3 round-by-round and use the chain rule” argument has a gap, and how a telescoping argument repairs it.
- **Quantitative comparisons (§6).** The per-client bound is a factor of K^2 tighter than (i) the whole-dataset bound $I(X; \tilde{\Delta} \mid P) \leq C^2/(2\sigma^2)$, and (ii) the per-client bound when each client’s clipped + noised update is released without averaging. Averaging *before* adding noise is therefore the active ingredient.

We deliberately avoid the chain “Gaussian mechanism $\Rightarrow (\epsilon, \delta)$ -DP $\Rightarrow$ per-record CMI” [e.g. Bun and Steinke, 2016, Steinke and Ullman, 2017, Bassily et al., 2018]. The direct route is shorter, gives explicit constants, and exposes *why* averaging helps: each client’s signal vector is scaled by $1/K$ inside the average, so its squared energy by $1/K^2$, and Lemma 1 converts squared signal energy into mutual information linearly.

Relation to prior work. The mutual-information memorization measure dates to Bassily et al. [2018], Russo and Zou [2016]; Brown et al. [2021] sharpened it to the conditional form $I(X; M \mid P)$ used here. DP-FedAvg analysis traditionally proceeds via (ϵ, δ) -DP accounting (RDP / moments accountant; see Abadi et al. 2016, Kairouz et al. 2021); we instead bound the CMI directly. Our proof for the multi-round case parallels classic adaptive composition arguments but operates on the CMI directly via a telescoping chain rule, which sidesteps the mismatch between marginal and conditional independence after seeing past releases.

2 Preliminaries

2.1 Memorization in the sense of Brown et al.

We use the framework of Brown et al. [2021, §1]. Fix a meta-distribution q over problem instances P , each itself a distribution on labeled examples. A learning algorithm A takes a dataset X of n i.i.d. examples from P and outputs $M = A(X)$. The (conditional) memorization is

$$\text{Mem}(A; q) := I(X; M \mid P), \quad (1)$$

computed in the joint model $P \sim q$, $X \sim P^{\otimes n}$, $M = A(X)$. Equation (1) measures how much the model reduces uncertainty about the realized training set beyond what is implied by knowing the data distribution.

2.2 Federated release model

We extend Brown et al. [2021] to a partitioned dataset. There are $K \geq 2$ clients. For $P \sim q$, client k holds local dataset X_k ; we write $X = (X_1, \dots, X_K)$ and X_{-k} for everything except client k . One round of DP-FedAvg proceeds as follows. Given the current iterate θ , each client computes a (possibly randomized) update vector

$$U_k = U_k(X_k, \theta, R_k) \in \mathbb{R}^d,$$

using fresh client randomness R_k . The server averages and adds Gaussian noise:

$$\tilde{\Delta} := \frac{1}{K} \sum_{k=1}^K U_k + Z, \quad Z \sim \mathcal{N}(0, \sigma^2 I_d), \quad (2)$$

with Z drawn independently of all data and client randomness. We make two assumptions.

Assumption 1 (Update clipping). For all clients k and all admissible (X_k, θ, R_k) , $\|U_k\|_2 \leq C$.

Assumption 2 (Conditional independence given P). Conditional on P , the client datasets $X_1, \dots, X_K$ are mutually independent. The client randomness $(R_1, \dots, R_K)$ is mutually independent and independent of $(X_1, \dots, X_K)$ and of P . The server noise Z is drawn fresh, independent of $(X, R_1, \dots, R_K, P)$.

Assumption 2 matches the Brown et al. generative model (i.i.d. examples from P) and is the natural extension to a partitioned dataset. It is the same independence model used implicitly in standard DP-FedAvg analyses.

3 A direct Gaussian-channel mutual-information bound

The only information-theoretic ingredient we need is the following standard bound; see, e.g., Cover and Thomas [2006, §9].

Lemma 1 (Bounded-energy MI through additive Gaussian noise). *Let S be an $\mathbb{R}^d$ -valued random variable with $\mathbb{E}\|S\|_2^2 < \infty$, and let $Z \sim \mathcal{N}(0, \sigma^2 I_d)$ be independent of S . Then*

$$I(S; S + Z) \leq \frac{1}{2\sigma^2} \text{tr Cov}(S) = \frac{1}{2\sigma^2} \mathbb{E}\|S - \mathbb{E}S\|_2^2 \leq \frac{1}{2\sigma^2} \mathbb{E}\|S\|_2^2. \quad (3)$$

Moreover, for any random variable W such that the joint pair (S, W) is independent of Z , the same bound holds conditionally:

$$I(S; S + Z \mid W) \leq \frac{1}{2\sigma^2} \mathbb{E}[\|S - \mathbb{E}[S \mid W]\|_2^2] \leq \frac{1}{2\sigma^2} \mathbb{E}\|S\|_2^2. \quad (4)$$

Proof. We use $I(S; S + Z) = h(S + Z) - h(S + Z \mid S) = h(S + Z) - h(Z)$, since conditional on S the random variable $S + Z$ has the same density as Z . Among all distributions on $\mathbb{R}^d$ with a fixed second-moment matrix, the Gaussian maximizes differential entropy [Cover and Thomas, 2006, Theorem 8.6.5], so

$$h(S + Z) \leq \frac{1}{2} \log((2\pi e)^d \det(\text{Cov}(S) + \sigma^2 I_d)),$$

where we used $\text{Cov}(S + Z) = \text{Cov}(S) + \sigma^2 I_d$ by independence of S and Z . Combining with $h(Z) = \frac{1}{2} \log((2\pi e)^d \det(\sigma^2 I_d))$ yields

$$I(S; S + Z) \leq \frac{1}{2} \log \det\left(I_d + \frac{1}{\sigma^2} \text{Cov}(S)\right).$$

The eigenvalue inequality $\log(1 + \lambda) \leq \lambda$ gives $\log \det(I_d + M) \leq \text{tr}(M)$ for PSD M , so

$$I(S; S + Z) \leq \frac{1}{2\sigma^2} \text{tr Cov}(S) = \frac{1}{2\sigma^2} \mathbb{E}\|S - \mathbb{E}S\|_2^2.$$

The final inequality in (3) is $\mathbb{E}\|S - \mathbb{E}S\|_2^2 = \mathbb{E}\|S\|_2^2 - \|\mathbb{E}S\|_2^2 \leq \mathbb{E}\|S\|_2^2$.

For the conditional version, condition on $W = w$. Since $Z \perp\!\!\!\perp (S, W)$, conditional on $W = w$ the noise Z is still $\mathcal{N}(0, \sigma^2 I_d)$ and independent of $S \mid W = w$. Apply (3) to the conditional distribution and integrate against P_W using the tower property and Jensen's inequality on the variance to obtain (4). $\square$

Remark 2 (Variance, not raw second moment). The first inequality in (3), $I \leq \frac{1}{2\sigma^2} \mathbb{E}\|S - \mathbb{E}S\|_2^2$, is sharper than the often-quoted $I \leq \frac{1}{2\sigma^2} \mathbb{E}\|S\|_2^2$. The clipping bound $\|S\| \leq C/K$ gives $\mathbb{E}\|S - \mathbb{E}S\|_2^2 \leq C^2/K^2$, but in many practical regimes $\mathbb{E}\|S - \mathbb{E}S\|_2^2$ is far smaller than C^2/K^2 (e.g. when client updates concentrate around a population mean), so (3) can be much tighter.

4 One-round per-client memorization bound

Theorem 3 (One-round). *Under Assumptions 1 and 2, the release $\tilde{\Delta}$ of (2) satisfies, for every $k \in \{1, \dots, K\}$,*

$$I(X_k; \tilde{\Delta} \mid P) \leq \frac{C^2}{2K^2\sigma^2}. \quad (5)$$

Proof. Fix k . Decompose

$$\tilde{\Delta} = \frac{1}{K}U_k + W_k + Z, \quad W_k := \frac{1}{K} \sum_{j \neq k} U_j.$$

Step 1: $X_k \perp\!\!\!\perp W_k \mid P$. By Assumption 2, conditional on P the pair (X_k, R_k) is independent of (X_{-k}, R_{-k}) ; since U_k is a measurable function of (X_k, R_k, θ) and W_k is a measurable function of (X_{-k}, R_{-k}, θ) (with θ constant in this round), this gives $(X_k, U_k) \perp\!\!\!\perp (W_k) \mid P$. Also, $Z \perp\!\!\!\perp (X, R_1, \dots, R_K, P)$, so $Z \perp\!\!\!\perp (X_k, U_k, W_k) \mid P$ as well.

Step 2: a chain-rule sandwich. Apply the chain rule for conditional MI two ways:

$$\begin{aligned} I(X_k; \tilde{\Delta}, W_k \mid P) &= I(X_k; W_k \mid P) + I(X_k; \tilde{\Delta} \mid P, W_k) \\ &= 0 + I(X_k; \tilde{\Delta} \mid P, W_k) \quad (\text{using Step 1}) \\ I(X_k; \tilde{\Delta}, W_k \mid P) &= I(X_k; \tilde{\Delta} \mid P) + I(X_k; W_k \mid P, \tilde{\Delta}). \end{aligned}$$

Since $I(X_k; W_k \mid P, \tilde{\Delta}) \geq 0$, the two equalities give

$$I(X_k; \tilde{\Delta} \mid P) \leq I(X_k; \tilde{\Delta} \mid P, W_k). \quad (6)$$

Step 3: subtract the known shift. Conditional on (P, W_k) , W_k is a fixed quantity, so $\tilde{\Delta} - W_k = \frac{1}{K}U_k + Z$ is a measurable bijection of $\tilde{\Delta}$. Hence

$$I(X_k; \tilde{\Delta} \mid P, W_k) = I(X_k; \frac{1}{K}U_k + Z \mid P, W_k). \quad (7)$$

By Step 1, $W_k \perp\!\!\!\perp (X_k, U_k, Z) \mid P$, so conditioning further on W_k does not change the joint distribution of $(X_k, \frac{1}{K}U_k + Z)$ given P :

$$I(X_k; \frac{1}{K}U_k + Z \mid P, W_k) = I(X_k; \frac{1}{K}U_k + Z \mid P). \quad (8)$$

Step 4: data processing and Gaussian channel. Conditional on P , we have the Markov chain $X_k \rightarrow U_k \rightarrow \frac{1}{K}U_k + Z$ (the second arrow uses fresh R_k and Z , independent of everything else given P). Conditional data processing gives

$$I(X_k; \frac{1}{K}U_k + Z \mid P) \leq I(U_k; \frac{1}{K}U_k + Z \mid P) = I(\frac{1}{K}U_k; \frac{1}{K}U_k + Z \mid P), \quad (9)$$

using that multiplication by K is a bijection. Apply the conditional Gaussian-channel bound (4) with $S = \frac{1}{K}U_k$ and conditioning variable P (recalling that $Z \perp\!\!\!\perp (S, P)$):

$$I(\frac{1}{K}U_k; \frac{1}{K}U_k + Z \mid P) \leq \frac{1}{2\sigma^2} \mathbb{E} \left[\left\| \frac{1}{K}U_k \right\|_2^2 \right] \leq \frac{C^2}{2K^2\sigma^2}, \quad (10)$$

where the last inequality uses Assumption 1. Chaining (6)–(10) yields (5). $\square$

Remark 4 (Sharpening via variance). Replacing the last inequality in (10) by $\frac{1}{2\sigma^2} \mathbb{E} \left\| \frac{1}{K}U_k - \mathbb{E}[\frac{1}{K}U_k \mid P] \right\|_2^2$ gives a strictly sharper bound when client updates concentrate (Remark 2); the form (5) uses only the worst-case clip.

5 Multi-round transcript bound

5.1 Multi-round release model

For $t = 0, 1, \dots, T-1$, at round t the server holds an iterate $\theta^{(t)}$ (a deterministic function of past releases¹). Each client k draws fresh $R_k^{(t)}$, produces a clipped update $U_k^{(t)} = U_k^{(t)}(X_k, \theta^{(t)}, R_k^{(t)})$ with $\|U_k^{(t)}\|_2 \leq C$, and the server releases

$$\tilde{\Delta}^{(t)} := \frac{1}{K} \sum_{k=1}^K U_k^{(t)} + Z^{(t)}, \quad Z^{(t)} \sim \mathcal{N}(0, \sigma_t^2 I_d), \quad (11)$$

with $\{Z^{(t)}\}_t$ and $\{R_k^{(t)}\}_{k,t}$ mutually independent and independent of (X, P) . Let $\mathcal{T} := (\tilde{\Delta}^{(0)}, \dots, \tilde{\Delta}^{(T-1)})$ and $\mathcal{H}^{(t)} := (\tilde{\Delta}^{(0)}, \dots, \tilde{\Delta}^{(t-1)})$, with $\mathcal{H}^{(0)} = \emptyset$.

¹If the server uses additional randomness to update $\theta^{(t)}$, independent of all data, simply augment the conditioning by that randomness; the proof is unchanged.

5.2 Why the one-round argument does not directly compose

A natural attempt is the chain rule $I(X_k; \mathcal{T} \mid P) = \sum_t I(X_k; \tilde{\Delta}^{(t)} \mid P, \mathcal{H}^{(t)})$ followed by the one-round proof at each t , with the conditioning augmented by $\mathcal{H}^{(t)}$. *This argument has a gap.* The one-round proof crucially used $X_k \perp\!\!\!\perp W_k \mid P$ (Step 1 of Theorem 3). After conditioning on $\mathcal{H}^{(t)}$, this fails: observing $\tilde{\Delta}^{(0)}, \dots, \tilde{\Delta}^{(t-1)}$ couples X_k with X_{-k} (an explaining-away effect), so in general $X_k \not\perp\!\!\!\perp W_k^{(t)} \mid P, \mathcal{H}^{(t)}$. The chain-rule sandwich (6) is no longer available with $W_k^{(t)}$ playing the role of W_k . We repair the argument by conditioning on the *full* X_{-k} instead of on $W_k^{(t)}$, exploiting that $X_k \perp\!\!\!\perp X_{-k} \mid P$ marginally.

5.3 Theorem and proof

Theorem 5 (Multi-round transcript memorization). *Assume Assumption 1 holds for every round and Assumption 2 holds with $(R_k^{(t)})_{k,t}$ and $(Z^{(t)})_t$ mutually independent fresh randomness. Then for every $k \in \{1, \dots, K\}$,*

$$I(X_k; \mathcal{T} \mid P) \leq \sum_{t=0}^{T-1} \frac{C^2}{2K^2\sigma_t^2}. \quad (12)$$

In particular, with $\sigma_t \equiv \sigma$, $I(X_k; \mathcal{T} \mid P) \leq TC^2/(2K^2\sigma^2)$. The same bound applies to any post-processed quantity, e.g. the final iterate $\theta^{(T)}$.

Proof. Step A: lift the conditioning to X_{-k} . By Assumption 2, $X_k \perp\!\!\!\perp X_{-k} \mid P$, so $I(X_k; X_{-k} \mid P) = 0$. Apply the chain rule:

$$\begin{aligned} I(X_k; \mathcal{T} \mid P) &\leq I(X_k; \mathcal{T}, X_{-k} \mid P) \\ &= I(X_k; X_{-k} \mid P) + I(X_k; \mathcal{T} \mid P, X_{-k}) \\ &= I(X_k; \mathcal{T} \mid P, X_{-k}). \end{aligned} \quad (13)$$

Step B: chain-rule expansion of the lifted MI. Applying the chain rule for conditional MI to $\mathcal{T} = (\tilde{\Delta}^{(0)}, \dots, \tilde{\Delta}^{(T-1)})$ given (P, X_{-k}) :

$$I(X_k; \mathcal{T} \mid P, X_{-k}) = \sum_{t=0}^{T-1} I(X_k; \tilde{\Delta}^{(t)} \mid P, X_{-k}, \mathcal{H}^{(t)}). \quad (14)$$

Step C: bound each round under full X_{-k} conditioning. Fix a round t and condition on $\mathcal{C}_t := (P, X_{-k}, \mathcal{H}^{(t)})$. The iterate $\theta^{(t)}$ is a deterministic function of $\mathcal{H}^{(t)}$, hence fixed. With X_{-k} and $\theta^{(t)}$ fixed,

$$W_k^{(t)} = \frac{1}{K} \sum_{j \neq k} U_j^{(t)}(X_j, \theta^{(t)}, R_j^{(t)})$$

is a measurable function of $\{R_j^{(t)}\}_{j \neq k}$, which is fresh randomness independent of $(X, P, \mathcal{H}^{(t)}, R_k^{(t)}, Z^{(t)})$. Therefore, given $\mathcal{C}_t$:

$$W_k^{(t)} \perp\!\!\!\perp X_k \mid \mathcal{C}_t, \quad Z^{(t)} \perp\!\!\!\perp (X_k, U_k^{(t)}, W_k^{(t)}) \mid \mathcal{C}_t. \quad (15)$$

Now repeat Steps 2–4 of the one-round proof with conditioning variable $\mathcal{C}_t$ in place of P :

$$\begin{aligned} I(X_k; \tilde{\Delta}^{(t)} \mid \mathcal{C}_t) &\stackrel{(*)}{\leq} I(X_k; \tilde{\Delta}^{(t)} \mid \mathcal{C}_t, W_k^{(t)}) \\ &= I\left(X_k; \frac{1}{K}U_k^{(t)} + Z^{(t)} \mid \mathcal{C}_t, W_k^{(t)}\right) \\ &= I\left(X_k; \frac{1}{K}U_k^{(t)} + Z^{(t)} \mid \mathcal{C}_t\right) \\ &\leq I\left(\frac{1}{K}U_k^{(t)}; \frac{1}{K}U_k^{(t)} + Z^{(t)} \mid \mathcal{C}_t\right) \\ &\leq \frac{1}{2\sigma_t^2} \mathbb{E}\left[\left\|\frac{1}{K}U_k^{(t)}\right\|_2^2\right] \leq \frac{C^2}{2K^2\sigma_t^2}, \end{aligned}$$

where $(*)$ uses (15) with the same chain-rule sandwich as (6); the second equality uses $W_k^{(t)} \perp\!\!\!\perp (X_k, U_k^{(t)}, Z^{(t)}) \mid \mathcal{C}_t$ (which is the conjunction of (15) and the freshness of $Z^{(t)}$); the data-processing step uses the Markov chain $X_k \rightarrow U_k^{(t)} \rightarrow \frac{1}{K}U_k^{(t)} + Z^{(t)}$ given $\mathcal{C}_t$; and the last inequality is Lemma 1 applied conditionally and Assumption 1.

Step D: combine. Plugging Step C into (14) and combining with (13) gives (12). The post-processing claim is the data-processing inequality applied to the deterministic map $\mathcal{T} \mapsto \theta^{(T)}$. $\square$

Remark 6 (Why the lift to X_{-k} is free). Step A uses only $I(X_k; X_{-k} \mid P) = 0$, the cleanest consequence of Assumption 2. Lifting the conditioning to X_{-k} (rather than to $W_k^{(t)}$ round-by-round) sidesteps the explaining-away dependence created by past releases entirely.

6 Discussion

6.1 Quantitative interpretation

For fixed clip C and noise σ , Theorem 3 gives a per-client memorization that decays as $O(1/K^2)$. On a Brown et al. task that requires the whole-dataset memorization $I(X; M \mid P) = \Omega(nd)$, the average per-client memorization is $\Omega(nd/K)$, and Theorem 3 therefore implies

$$\sigma^2 = \Omega\left(\frac{C^2}{K \cdot nd}\right)$$

as a necessary noise scale for one-round release on such tasks. The T -round version of the same calculation gives $T \cdot C^2/(2K^2\sigma^2) \geq \Omega(nd/K)$, i.e., $\sigma^2 = \Omega(TC^2/(K \cdot nd))$.

6.2 Comparison: averaging vs. no averaging

If each client's clipped update were instead released with independent Gaussian noise of the same scale, Lemma 1 gives $I(X_k; U_k + Z_k \mid P) \leq C^2/(2\sigma^2)$. Theorem 3 is a factor of K^2 tighter. Averaging *before* adding noise at the same per-coordinate scale therefore strictly improves per-client memorization by K^2 . The improvement is information-theoretic; it is not a privacy-accounting artifact.

6.3 Comparison: whole-dataset bound

Applying Lemma 1 directly to $S = \frac{1}{K} \sum_k U_k$ and using the triangle inequality $\|S\| \leq C$ gives the whole-dataset bound $I(X; \tilde{\Delta} \mid P) \leq C^2/(2\sigma^2)$. Theorem 3 is sharper for individual clients by a factor of K^2 , because each client's signal is scaled by $1/K$ within the average even though the average's overall norm need not shrink.

6.4 From per-client to per-record memorization

If client k 's data is $X_k = (X_{k,1}, \dots, X_{k,n_k})$, the data-processing inequality gives $I(X_{k,i}; M \mid P) \leq I(X_k; M \mid P) \leq C^2/(2K^2\sigma^2)$ (one-round) and the analogous transcript bound. Per-record memorization is therefore at most the per-client bound. The bound makes no assumption about how U_k is computed from X_k , so it applies whether U_k is a one-step gradient, a multi-step local update, a model delta, or a more complex client-side optimization output.

6.5 What this paper does not say

We make no claim about plain (non-private) FedAvg, where the released signal is $\Delta = \frac{1}{K} \sum_k U_k$ with no noise. The mutual information $I(X_k; \Delta \mid P)$ can be $\Omega(1)$ —or even saturate $H(X_k \mid P)$ —without further structural assumptions, since Δ is a deterministic function of (X, R) and high-dimensional models do not offer an inherent contraction. Establishing strong data-processing-style contractions for vanilla averaging in realistic regimes is an interesting open direction.

Acknowledgments

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